# Yijia Li

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*Engine-proficient **Game Designer** blending technical expertise with creative vision to deliver exceptional gaming experiences*

## EDUCATION

**Duke University**, **Durham, NC** | 5/2027 (Expected)
Game Development, Design & Innovation (M.Eng.)
**Boston University**, **Boston, MA** | 5/2025
Computer Science (BA)

## SKILLS

**Engines & Tools**: Unreal Engine, Unity, Perforce, Jira, GitHub
**Languages**: C++, C#, Java, Python, JavaScript
**Soft Skills**: Leadership, Critical Thinking, Self-Awareness, Player-Centered Thinking

## PROFESSIONAL EXPERIENCE

**Brokencigs, LLC** | **Inkression** **(Steam) | 3D Narrative Exploration Game** **6/2024 – 8/2024**
*Game Design Intern* **Hybrid (New York, NY)**

- Designed two core interactive systems, a 2D/3D spatial alignment Graffiti system and an audio sampling Voice Recorder mini-game, which connected discovery-based mechanics directly to themes of urban memory and overlooked city details, using visual and auditory storytelling to deepen player immersion.
- **Graffiti Mechanic**: Designed **2D-to-3D environmental puzzles** requiring players to analyze urban geometry and lighting to match graffiti tags with specific NPCs, unlocking character backstories and questlines.
- **Voice Recorder Mechanic**: Engineered an **interactive audio sampling mini-game** where players locate ambient sounds, calibrate input gain, and record environmental audio to build unique musical compositions for a street musician.

## SELECTED PROJECTS EXPERIENCE

**Gravity** **(Graduate Capstone Project) | 3D Puzzle Platforming | Team Size: 18 | Unreal** **8/2025 – Now**
*Lead Designer, Programmer, Producer* **Durham, NC**

- Structured Chapter I into **6 escalating beats** via a "**Teach–Test–Twist**" framework, mapping mechanic rollouts directly to level layout complexity. Paced a **30-minute interest curve** from onboarding to core traversal mastery, ending with a high-stakes, low-gravity timed set piece featuring dynamic geometry and hazards.
- Built a C++/Blueprint airlock room that **masks level streaming** through state-driven doors and orientation-preserving teleportation, enabling seamless transitions between gameplay spaces.
- Engineered a **deterministic destruction framework** for Chapter I's climactic set piece, synchronizing Chaos destruction, moving environments, and player control to deliver a cinematic finale with controlled gameplay outcomes.

**Arcade Tank Fight** **| Top-Down Shooter | Solo Project | Unreal** **2/2026 – 5/2026**
*Game Designer, Programmer* **Durham, NC**

- Built a **Gameplay Ability System (GAS) framework** for tank combat, using abilities, attributes, and simple VFX to manage firing restrictions, movement modifiers, and power-up states.
- Engineered a **per-class object pooling subsystem** for three projectile behaviors (straight, guided, bouncing), providing more gameplay options for players.

**Sneaking Out** **| 3D Stealth | Team Size: 5 | Unreal** **8/2025 – 12/2025**
*Game Developer* **Durham, NC**

- Implemented character locomotion mechanics (sprint, crouch, jump) utilizing Unreal Engine's **Enhanced Input System** integrated with a **Finite State Machine (FSM)** to gate transitions for stealth pacing and control fluidity.
- Designed and implemented a throw-to-distract mechanic with **spline-based** trajectory prediction, **animation montages**, **physics-based impulses**, and inventory integration, enabling precise and intuitive player control.